

SUSHANT LOKHANDE

San Jose, CA

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PROFESSIONAL SUMMARY

Software Engineer experienced in backend development, distributed systems, applied AI, and machine learning. Strong in Python, C++, Go, APIs, real-time systems, PyTorch, cloud infrastructure, and building reliable software from experimentation through production.

TECHNICAL SKILLS

Languages: Python, C++, Go, Java, TypeScript, JavaScript, Rust, SQL
Backend & Systems: FastAPI, REST APIs, gRPC, Distributed Systems, Event-Driven Systems, WebSockets, WebRTC
AI & Machine Learning: PyTorch, scikit-learn, CNNs, Embeddings, LLM APIs, RAG, Tool Calling, Model Evaluation
Data: PostgreSQL, Redis, Kafka, Vector Search, SQL
Cloud & Infrastructure: AWS, Docker, Kubernetes, Linux, CI/CD, GitHub Actions
Observability: OpenTelemetry, Prometheus, Grafana, Distributed Tracing

WORK EXPERIENCE

San Jose State University | Graduate Research Assistant, Computer Science Aug 2025 to May 2026

- Improved malware classification accuracy to **80.2%**, outperforming the best individual model by **2.4 percentage points**, by combining complementary machine learning models across **17,000 samples and 17 malware families**.
- Reduced experiment setup time by **55%** by building Python automation for configuration, distributed execution, telemetry collection, and reproducible analysis across approximately **300 containerized workers**.
- Executed more than **1,200 controlled experiments** to evaluate system performance, reliability, and adaptive behavior under changing workload and network conditions.
- Improved model efficiency by reducing a feature representation from more than **6,000 features to 103** while achieving **77.8% classification accuracy** through systematic feature selection and evaluation.

Turito | Software Engineering Intern, Real-Time Systems Jun 2022 to Oct 2022

- Reduced latency-sensitive CPU utilization by **22%** by profiling real-time processing paths and optimizing memory allocation, payload copying, routing lookups, shared-state access, and instrumentation overhead.
- Improved connection reliability under controlled network disruption by reducing failures from **3.1% to 1.8%** through stronger routing, timeout handling, retries, and recovery logic.
- Improved reliability of distributed backend workflows by developing asynchronous services, validation, persistent state, bounded retries, and automated failure recovery.
- Reduced production diagnosis time by **35%** by adding structured logging, distributed tracing, service metrics, dashboards, and request-level diagnostics.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Validated distributed backend services at up to **25K lightweight concurrent sessions** by building Rust and Python tooling for load generation, connection churn, failure injection, and recovery testing.
- Improved equivalent-load throughput by **31%** by identifying serialized WebSocket writes, shared-state contention, worker starvation, and inefficient resource cleanup.
- Reduced investigation time by **40%** by building internal tooling that consolidated execution results, performance metrics, configuration state, failures, and historical runs.

RELEVANT PROJECTS

Reel Rank | Recommendation & Retrieval System — PyTorch, C++, FastAPI, React

- Built a full-stack recommendation system combining neural retrieval, a C++ HNSW approximate-nearest-neighbor index, learned ranking, FastAPI services, and a React/TypeScript interface.
- Improved recommendations for sparse-user scenarios by separating candidate generation from ranking and incorporating metadata alongside learned user-item representations.

Malware Vision | Machine Learning Classification System — Python, PyTorch, CNNs

- Built a malware classification pipeline that transformed binary samples into image representations and evaluated handcrafted features, pretrained neural networks, custom CNNs, and ensemble models across **17 malware families**.
- Achieved **80.2% test accuracy** with a soft-voting ensemble, improving over the strongest individual model by **2.4 percentage points** through complementary model combination and systematic evaluation.

EDUCATION

San Jose State University | M.S. in Computer Science, GPA: 3.8/4.0 Aug 2024 to May 2026

University of Mumbai | B.Tech. in Computer Engineering, GPA: 3.9/4.0 Jun 2019 to May 2023